Week 4 Assessment

Advanced Video Processing with MediaPipe

Duration: 3 Hours **Total Marks:** 100

Instructions

• Use Python, OpenCV, and MediaPipe only.

• Use either a webcam or recorded video, as specified in the question.

• Submit the complete Python program, output video/screenshots, and a brief explanation of the algorithm.

• Students may use suitable threshold values and clearly document them.

Question 1 — Gesture-Controlled Smart Video Zoom In / Zoom Out (20 Marks)

Develop a real-time application using MediaPipe Hand Tracking to control video zoom.

Requirements

1. Detect the hand and display landmarks.
2. Calculate thumb-tip and index-fingertip distance.
3. Increase distance → Zoom In; decrease distance → Zoom Out.
4. Implement zoom using ROI cropping and resizing.
5. Display the current zoom level.
6. Keep the zoomed frame within video boundaries.

Expected Processing Flow

```
Video Input
  ↓
MediaPipe Hand Tracking
  ↓
Thumb + Index Fingertip
  ↓
Calculate Distance
  ↓
Determine Zoom Factor
  ↓
Dynamic ROI → Crop + Resize
  ↓
Zoomed Video Output
```

Question 2 — Virtual Air Drawing Board (20 Marks)

Develop an interactive drawing application using MediaPipe Hand Tracking.

Requirements

1. Detect the hand and index fingertip.
2. Use the fingertip as the drawing coordinate.
3. Create a persistent drawing canvas.
4. Draw continuously as the finger moves.
5. Display drawing with the live video.
6. Add one gesture-based control for clearing, changing mode, or starting/stopping drawing.

Expected Processing Flow

```
Video Input
  ↓
MediaPipe Hand Tracking
  ↓
Index Fingertip Position
  ↓
Store Previous Position
  ↓
Draw on Canvas
  ↓
Combine Canvas + Video
  ↓
Interactive Output
```

Question 3 — Pose-Based Fall Detection (20 Marks)

Develop a single-person fall detection system using MediaPipe Pose and recorded video.

Requirements

1. Detect pose landmarks.
2. Extract visible landmark coordinates.
3. Calculate a body bounding region.
4. Analyze width-to-height ratio and/or body orientation.
5. Use consecutive frames to reduce single-frame false detection.
6. Display NORMAL or FALL DETECTED.
7. Draw landmarks and the bounding region.

Note: Use MediaPipe Pose only; this question is intended for a single-person video.

Expected Processing Flow

```
Recorded Video
  ↓
MediaPipe Pose
  ↓
33 Pose Landmarks
  ↓
Body Geometry Analysis
  ↓
Posture / Orientation
  ↓
Temporal Frame Counter
  ↓
NORMAL / FALL DETECTED
```

Question 4 — Driver Drowsiness and Yawn Detection (20 Marks)

Develop a driver-monitoring application using MediaPipe Face Landmarks.

Requirements

1. Detect facial landmarks from a driver video.
2. Determine eyes open/closed.
3. Detect prolonged eye closure using consecutive frames.
4. Determine mouth open/closed.
5. Detect a possible yawn using mouth opening and duration.
6. Display suitable alerts: NORMAL, EYES CLOSED, YAWN DETECTED, or DROWSINESS ALERT.

Expected Processing Flow

```
Driver Video
  ↓
MediaPipe Face Landmarks
  ↓
Eye Analysis + Mouth Analysis
  ↓
Open / Closed Measurements
  ↓
Closure Time + Yawn Duration
  ↓
Drowsiness Decision
  ↓
Driver Alert
```

Question 5 — Person Background Removal and Virtual Background (20 Marks)

Develop a video application using MediaPipe Image Segmentation.

Requirements

1. Read webcam or recorded video.
2. Generate a person segmentation mask.
3. Display the segmentation mask.
4. Remove or replace the original background.
5. Use an image as a virtual background.
6. Combine the foreground person with the new background.
7. Display original and final output.

Expected Processing Flow

```
Video Input
  ↓
MediaPipe Image Segmentation
  ↓
Person Segmentation Mask
  ↓
Separate Foreground
  + Virtual Background
  ↓
Video Compositing
  ↓
Final Output
```

Assessment Summary

Question	Project	MediaPipe Component	Marks
Q1	Gesture-Controlled Smart Video Zoom	Hands	20
Q2	Virtual Air Drawing Board	Hands	20
Q3	Pose-Based Fall Detection	Pose	20
Q4	Driver Drowsiness and Yawn Detection	Face Landmarks	20
Q5	Person Background Removal and Virtual Background	Image Segmentation	20
Total			**100**

Coverage: Hand interaction → gesture-based control → pose analysis → face analysis → video segmentation and compositing.